THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 9: **Propositions and their Converses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A triangle has sides 39 cm, 80 cm and 89 cm. Using the converse of the Baudhāyana–Pythagoras theorem, the triangle is: 1

(a) equilateral **(b)** isosceles **(c)** right-angled **(d)** not right-angled

2. Which statement is true for every integer n? 1

(a) $2n+1$ is even **(b)** $3n$ is odd **(c)** $n+2$ is even **(d)** $2n+1$ is odd

3. The sum of the 3 consecutive integers starting from 41 is: 1

(a) 127 **(b)** 129 **(c)** 126 **(d)** 123

4. The number of integers from 1 to 200 that have an odd number of factors is: 1

(a) 16 **(b)** 15 **(c)** 13 **(d)** 14

5. The converse of "If $x=3$, then $x^2=9$" is false because: 1

(a) 6 is even but not a multiple of 4

(b) two angles of 60° in an equilateral triangle are equal but not vertically opposite

(c) a 4 cm by 6 cm rectangle is not a square

(d) $x=-3$ gives $x^2=9$ but $x\neq 3$

6. Which of the following is a counterexample to the statement "The difference of two primes is prime"? 1

(a) 5 and 2 **(b)** 11 and 3

(c) 13 and 2 **(d)** 7 and 2

7. For the expression n^2-n+59, the smallest natural number n for which the value is NOT prime is: 1

(a) 3 **(b)** 4 **(c)** 2 **(d)** 5

8. The number of factors of 210 is: 1

(a) 18 **(b)** 16 **(c)** 15 **(d)** 17

9. A triangle has sides 45 cm, 60 cm and 76 cm. Using the converse of the Baudhāyana–Pythagoras theorem, the triangle is: 1

(a) equilateral **(b)** isosceles **(c)** not right-angled **(d)** right-angled

10. Which statement is true for every integer n? 1

(a) n^3+n is odd **(b)** n^3-n is odd

(c) n^3 is even **(d)** n^3-n is divisible by 2

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Is the number 169 a perfect square? Decide by counting its factors. 2

12. Write the converse of "If a number is divisible by 9, then the sum of its digits is divisible by 9" and test it. 2

13. Find the smallest natural number n for which $n^2 - n + 5$ is not prime, and the value obtained. 2

14. Is the number 200 a perfect square? Decide by counting its factors. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the converse of "If a triangle is equilateral, then all its angles are 60°". Is the converse true? Justify. 3

16. Show that the statement "If a number is divisible by 4 and by 6, then it is divisible by 24" is false by giving a counterexample. 3

17. If n is a multiple of 28, is n a multiple of 7? Write the converse and give a counterexample if false. 3

OR

Write the converse of "If a number is divisible by 9, then the sum of its digits is divisible by 9". Is the converse true? Justify. 3

18. Show that the statement "The product of two irrational numbers is irrational" is false by giving a counterexample. 3

OR

If n is a multiple of 45, is n a multiple of 15? Write the converse and give a counterexample if false. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that the sum of 5 consecutive even numbers starting at $2n$ is a multiple of 10. Check with $n = 12$. 5

OR

Take the two-digit number 38. Subtract the sum of its digits and show that the result is divisible by 9. Prove the same for any two-digit number. 5

20. Prove that the sum of 9 consecutive even numbers starting at $2n$ is a multiple of 18. Check with $n = 2$. 5

OR

Take the two-digit number 71. Subtract the sum of its digits and show that the result is divisible by 9. Prove the same for any two-digit number. 5

END OF QUESTION PAPER — Total: 40 Marks